MMP 1
Serie 6
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Viel rechnen ᕦ(ò_óˇ)ᕤ

1 MC

(A)

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3

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} \tag{3.1}$$

3.a

$z = y - x$

$$\begin{aligned} e^{2xy-y^2} &= e^{x^2} e^{-z^2} \\ &= e^{x^2} \sum_{n=0}^{\infty} \frac{\partial^n}{\partial z_0^n}\left(e^{-z_0^2}\right) \frac{(z-z_0)^n}{n!}, \qquad z_0 = -x \\ &= e^{x^2} \sum_{n=0}^{\infty} (-1)^n e^{-x^2} \frac{H_n(z_0)}{n!} y^n \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{H_n(-x)}{n!} y^n \\ &= \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} y^n \end{aligned} \tag{3.2}$$

3.b

3.b.i

$$\begin{aligned} & H_{n+2}(x) - 2xH_{n+1}(x) + 2(n+1)H_n(x) = 0 \\ \Leftrightarrow\ & \frac{d^{n+2}}{dx^{n+2}} e^{-x^2} + 2x \frac{d^{n+1}}{dx^{n+1}} e^{-x^2} + 2(n+1) \frac{d^n}{dx^n} e^{-x^2} = 0 \end{aligned} \tag{3.3}$$

$$\begin{aligned} & \frac{\partial}{\partial x} e^{-x^2} = -2xe^{x^2} \Rightarrow \left(\frac{\partial}{\partial x} e^{-x^2} + 2xe^{-x^2}\right) = 0 \\ \Rightarrow\ & \frac{\partial^{n+1}}{\partial x^{n+1}} \left(\frac{\partial}{\partial x} e^{-x^2} + 2xe^{-x^2}\right) = \frac{\partial^{n+2}}{\partial x^{n+2}} e^{-x^2} + 2 \frac{\partial^{n+1}}{\partial x^{n+1}} \left(xe^{-x^2}\right) = 0 \end{aligned} \tag{3.4}$$

$$\begin{aligned} & \frac{\partial^n}{\partial x^n}(g(x)f(x)) = \sum_{j=0}^{n} \binom{n}{j} g^{(j)} f^{(n-j)} \qquad g(x) = x, \quad f(x)e^{-x^2} \\ & \Rightarrow \sum_{j=0}^{1} \binom{n}{j} x^{(n)} \frac{\partial^{n-j}}{\partial x^{n-j}} \left(e^{-x^2}\right) = x \frac{\partial^n}{\partial x^n} e^{-x^2} + n \frac{\partial^{n-1}}{\partial x^{n-1}} e^{-x^2} \end{aligned} \tag{3.5}$$

$$\Rightarrow H_{n+2}(x) - 2xH_{n+1}(x) + 2(n+1)H_n(x) = 0 \tag{3.6}$$

3.b.ii

$$H_n'(x) - 2nH_{n-1}(x) = 0 \tag{3.7}$$

$$H_n'(x) = 2x(-1)^n e^{x^2} \frac{\partial^n}{\partial x^n} e^{-x^2} + (-1)^n e^{x^2} \frac{\partial^{n+1}}{\partial x^{n+1}} e^{-x^2} = 2xH_n(x) - H_{n+1}(x) \tag{3.8}$$

$$\begin{aligned} H_n''(x) &= (2+4x^2)(-1)^n e^{x^2} \frac{\partial^n}{\partial x^2} e^{-x^2} + 2x(-1)^n e^{x^2} \frac{\partial^{n+1}}{\partial x^{n+1}} e^{-x^2} + 2x(-1)^n e^{x^2} \frac{\partial^{n+1}}{\partial x^{n+1}} e^{-x^2} + H_{n+2}(x) \\ &= (2+4x^2)H_n(x) - 4xH_{n+1}(x) + H_{n+2}(x) \end{aligned} \tag{3.9}$$

using (3.6)

$$2nH_{n-1} = 2xH_n - H_{n+1}(x) \tag{3.10}$$

with (3.7) and (3.8)

$$\Rightarrow H_n'(x) - 2nH_{n-1}(x) = 0 \tag{3.11}$$

3.b.iii

with (3.9), (3.8), (3.7), (3.6)

$$\begin{aligned} H_n'' - 2xH_n' + 2nH_n &= (2+4x^2)H_n - 4xH_{n+1} + H_{n+2} - 4x^2H_n + 2xH_{n+1} - H_{n+2} + 2xH_{n+1} - 2H_n \\ &= 0 \end{aligned} \tag{3.12}$$

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4.a

$$\begin{aligned} \frac{\alpha^2}{z^2}J_\alpha &= \frac{\alpha^2}{z^2}\sum_{l=0}^{\infty}\frac{(-1)^l}{l!\Gamma(l+\alpha+1)}\left(\frac{z}{2}\right)^{2l+\alpha} \\ &= \sum_{l=0}^{\infty} -\frac{(-1)^l\alpha^2}{l!\Gamma(l+\alpha+1)}\left(\frac{z}{2}\right)^{2l+\alpha-2}\frac{1}{4} \end{aligned} \tag{4.1}$$

$$\frac{J_\alpha'}{z} = \sum_{l=0}^{\infty}\frac{(-1)^l(2l+\alpha)}{l!\Gamma(l+\alpha+1)}\left(\frac{z}{2}\right)^{2l+\alpha-2}\frac{1}{4} \tag{4.2}$$

$$J_\alpha'' = \sum_{l=0}^{\infty}\frac{(-1)^l(2l+\alpha)(2l+\alpha-1)}{l!\Gamma(l+\alpha+1)}\left(\frac{z}{2}\right)^{2l+\alpha-2}\frac{1}{4} \tag{4.3}$$

$$\Rightarrow (2l+\alpha) + (2l+\alpha)(2l+\alpha-1) - \alpha^2 = (2l+\alpha)^2 - \alpha^2 = 4l(l+\alpha) \tag{4.4}$$

together

$$\begin{aligned} \Rightarrow J_\alpha'' + \frac{1}{z}J_\alpha' - \frac{\alpha}{z^2}J_\alpha &= \sum_{l=0}^{\infty}\frac{(-1)^l 4l(l+\alpha)}{4l!\Gamma(l+\alpha+1)}\left(\frac{z}{2}\right)^{2l+\alpha-2} \\ &= \sum_{l=0}^{\infty}\frac{(-1)^l}{(l-1)!\Gamma((l-1)+\alpha+1)}\left(\frac{z}{2}\right)^{2(l-1)+\alpha} \\ &= \sum_{k=0}^{\infty}\frac{-(-1)^k}{k!\Gamma(k+\alpha+1)}\left(\frac{z}{2}\right)^{2k+\alpha} = J_\alpha \end{aligned} \tag{4.5}$$

□

4.b

$$W(f,g) = \det\left(\begin{pmatrix} f & g \\ f' & g' \end{pmatrix}\right) = fg' - f'g \tag{4.6}$$

$$W'(f,g) = fg'' - f''g \tag{4.7}$$

if g is a solution

$$g'' + \frac{1}{z}g' + \left(1 - \frac{n^2}{z^2}\right)g = 0 \tag{4.8}$$

$$\begin{aligned} \Rightarrow W'(f,g) &= -\frac{1}{z}g'f - \left(1-\frac{n^2}{z^2}\right)gf + \frac{1}{z}gf' + \left(1-\frac{n^2}{z^2}\right)gf \\ &= -\frac{1}{z}W(f,g) \end{aligned} \tag{4.9}$$

$$\int \frac{1}{W}dW = -\int \frac{1}{z}dz \tag{4.10}$$

$$\Rightarrow W(f,g) = \frac{1}{z}C \tag{4.11}$$

$$\lim_{z\to\infty} W(f,g) = 0 \tag{4.12}$$

$\Rightarrow f, g$ sind linear abhängig

4.c

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$$J_n'(z) = \frac{1}{2\pi}\int_0^{2\pi} i\sin(\theta)e^{-in\theta}e^{iz\sin(\theta)}d\theta \tag{5.1}$$

$$J_n''(z) = \frac{1}{2\pi}\int_0^{2\pi} -\sin^2(\theta)e^{-in\theta}e^{iz\sin(\theta)}d\theta \tag{5.2}$$

$$\begin{aligned}&\Rightarrow \int_0^{2\pi}(-z^2\sin^2(\theta) + zi\sin(\theta) + z^2 + n^2)e^{-in\theta}e^{iz\sin(\theta)}d\theta \\ &= \int_0^{2\pi}(z^2\cos^2(\theta) + zi\sin(\theta) + n^2)e^{-in\theta}e^{iz\sin(\theta)}d\theta \\ &...?\end{aligned} \tag{5.3}$$

6

6.a

$$J_\alpha'(z) + \frac{\alpha}{z}J_\alpha(z) = J_{\alpha-1}(z) \Rightarrow J_\alpha^l\left(\frac{2l+\alpha+\alpha}{z}\right) = J_\alpha^l(l+\alpha)\frac{2}{z} \tag{6.1}$$

□

6.b

$$\begin{aligned}J_\alpha'(z) - \frac{\alpha}{z}J_\alpha(z) &= \sum_{l=0}^{\infty}\frac{(-1)^l}{l!\Gamma(l+\alpha+1)}\left(\frac{z}{2}\right)^{2l+\alpha}\frac{2l+\alpha-\alpha}{z} \\ &= \sum_{l=0}^{\infty}\frac{(-1)^l}{(l-1)!\Gamma((l-1)+(\alpha+1)+1)}\left(\frac{z}{2}\right)^{2l+\alpha+1-2} \\ &= -\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\Gamma(k+\alpha+2)}\left(\frac{z}{2}\right)^{2k+\alpha+1} = -J_{\alpha+1}(z)\end{aligned} \tag{6.2}$$

6.c

$$\begin{aligned}J_{\alpha-1}(z) + J_{\alpha+1}(z) &= J_\alpha'(z) + \frac{\alpha}{z}J_\alpha(z) + J_{\alpha+1}(z) \\ &= -J_{\alpha+1}(z) + \frac{\alpha}{z}J_\alpha(z) + \frac{\alpha}{z}J_\alpha(Z) + J_{\alpha+1}(z) = \frac{2\alpha}{z}J_\alpha(z)\end{aligned} \tag{6.3}$$

6.d

$$J_{\alpha-1}(z) - J_{\alpha+1}(z) = J_\alpha' + \frac{\alpha}{z}J_\alpha + J_\alpha' - \frac{\alpha}{z}J_\alpha = 2J_\alpha'(z) \tag{6.4}$$